Stock Market Prediction Web App using LSTM Model and Facebook Prophet

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2024 International Conference on Smart Technologies for Sustainable Development Goals (ICSTSDG) | 979-8-3315-1628-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICSTSDG61998.2024.11026584

***Abstract*—The Prophet and LSTM (Long Short-Term Memory) models combined powers are presented in the project as comprehensive method of stock price prediction through a single Python script. The script preprocesses historical stock data for both models using information received from yfinance.Time series forecasting is done with the Prophet model, which projects stock prices for the next 365 days and uses Plotly to visualise the results. The script divides the information into sets for testing and training, scales the information, then trains an LSTM model on the training set simultaneously. For a certain time period, the closing prices are projected using the LSTM model, and the forecasts are assessed using RMSE, or root mean squared error.Using Plotly, The script further illustrates the difference between real and forecasted stock values.This research offers an integrated approach to forecast stock prices by combining Sequential data processing's capabilities of the LSTM model with the forecasting skills of Prophet. It draws attention to how well these models complement one another in explaining various facets of stock price movements and emphasises how important it is to use machine learning predictions into a more comprehensive approach to investing analysis.To sum up, this study demonstrates how Prophet and LSTM models may be used in concert to predict stock prices, offering important new perspectives on the potential use of these models in financial forecasting. It emphasises how important it is to make well- informed decisions and seek professional advice when using machine learning models for investing.**

Keywords*—Streamlit, Pandas, NumPy, Keras (TensorFlow), yfinance, Prophet (Facebook),Plotly, LSTM, Slider(UI interaction), Data caching.

I. Introduction

Accurate stock price prediction is essential for traders and investors hoping to make educated decisions in the dynamic world of financial markets.We describe an advanced web app for stock market prediction that uses Facebook Prophet and Long Short-Term Memory models to fulfil this requirement.This cutting-edge technology gives users the ability to accurately predict stock market movements and obtain insightful knowledge about the workings of the market.Our initiative is driven by the realisation that investors have difficulties when attempting to navigate the volatile nature of stock markets.Conventional forecasting techniques frequently fail to capture the volatility and complexity included in financial data.As a result, we worked to create a system that would use cutting-edge machine learning algorithms' predictive powers to produce precise and useful forecasts. Our Stock Market Prediction Web App's main goal is to give users a dependable and intuitive stock price forecasting platform.Our goal is to offer a thorough analysis of the stock market analysis that appeals to both long-term investors and short-term traders by combining Facebook Prophet and LSTM models.Our mission is to provide people with practical insights so they may make wise decisions and optimise their stock market results.Our platform's primary characteristics are its interactive user interface, performance evaluation measures, and innovative forecasting methodologies. Comprehensive forecasting is made possible by the combination of Facebook Prophet and LSTM models, which encompass both Long and short-term trends in the stock pricing.With the use of the web application's user-friendly interface, users may enter stock symbols, choose prediction parameters, and view predicted trends using interactive graphs and charts. Users are able to evaluate the forecasts' dependability by utilising critical performance measures including the forecast accuracy is assessed using the root mean squared error made by Facebook Prophet and LSTM models.In summary, our Stock Market Prediction Web App gives users a strong tool for making educated judgements in the fast-paced financial markets of today, marking a substantial improvement in the domain of stock market prediction. Our goal is to provide customers with the necessary information to successfully traverse the intricacy of the stock market and reach the financial goals by utilising cutting-edge machine learning techniques with an intuitive interface design. Regardless of your level of experience, our platform offers you the tools and guidance you need to be successful in the stock trading industry.

II. Literature Study

Numerous techniques and strategies aimed at improving the reliability and accuracy of stock price prediction are included in the overview of the literature on ML approaches for stock market prediction.

A thorough examination outlining the advantages and disadvantages of various machine learning techniques for forecasting the stock market was carried out by Ashish Sharma et al. (2017). Ze Zhang et al. (2017) demonstrated the flexibility and efficacy of neural network architectures by proposing a novel method for the very small span prediction of stock market opening prices using a self-adapting variant PSO-Elman neural network version . In the meantime, Nonita Sharma and Akanksha Juneja (2017) presented a technique that uses LSboost to combine arbitrary forest estimaes using LSboost to predicting stock market indices, showcasing the potential of ensemble methods in enhancing prediction accuracy. The use of social media mining technologies to help predict stock market values was investigated by Yaojun Wang and Yaoqing Wang (2016), demonstrating the increasing significance of alternative data sources including sentiment analysis of social media data. In order to predict stock market values, Mustain Billah et al. (2016) suggested a better

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neural network training algorithm, underscoring continuous efforts to improve prediction accuracy and the learning process. Sneh Kalra and Jay Shankar Prasad (2019) investigated the usefulness of news sentiment analysis for stock market prediction and offered insight into how sentiment analysis works to extract market sentiments from news stories.

An thorough analysis of the literature on the application of ANNs to stock market forecasting was carried out by Muhammad Firdaus et al. (Year), who also included insightful information about how ANN approaches have evolved in the field of financial forecasting. Tanapon Tantisripreecha and Nuanwan Soonthomphisaj (2018) demonstrated the integration of topic modelling techniques with machine learning algorithms by proposing a model for predicting stock market movements based on Latent Dirichlet Allocation (LDA) and online learning. The possibility of sentiment analysis to increase prediction accuracy was demonstrated by Zhaoxia Wang et al.'s (2018) investigation into the integration of sentiment analysis and predictions about the stock market based on news and social media opinions.

Last but not least, Saloni Mohan et al. (2019) demonstrated how to use news sentiment analysis to predict stock prices, emphasising the significance of integrating sentiment analysis of news articles into predictive models. When taken as a whole, these studies help to enhance stock market forecasting techniques and pave the way for more accurate and reliable financial forecasting instruments.

III. Architecture & Methodology

Facebook Prophet and Long Short-Term Memory (LSTM) models, two distinct methods for stock price forecasting, are combined in the online stock market prediction application. First, historical stock market data is retrieved via the Yahoo Finance API and preprocessed to guarantee consistency and fill in any missing information . After employing exploratory data analysis (EDA) approaches to uncover data trends, feature engineering is used to extract pertinent characteristics like moving averages. After that, Keras is used to build LSTM models, they are subsequently assessed using Root Mean Squared Error (RMSE) after being trained on some historical data . Facebook Prophet models are also created to identify trends and seasonality in the data. Through an interactive online interface made with Streamlit, users may modify forecast periods and input stock symbols for the application.Users are aided in forecast interpretation and decision-making by visualisations of past data, estimatedpricing, and assessment measures.

A. LSTM Model

The Long Short-Term Memory (LSTM) model, a variant of recurrent neural networks (RNNs), is a crucial part of the realtime stock market prediction service. The use of LSTM models in stock price forecasting is highly advantageous since they are adept at spotting long-term trends and sequential dependencies in time-series data. Typically, the architecture of an LSTM model is composed of multiple LSTM layers, which are succeeded by dense layers for prediction. The algorithm is trained with historical stock market data to predict future price changes. Important steps in employing the LSTM model include feature engineering, training, data preprocessing, and model evaluation. Evaluation measures such as Root Mean Squared Error (RMSE) are commonly employed to assess the precision of forecasts.The web tool incorporates the LSTM model to produce projections groundedin historical data, offering users insightful perspectives into possible patterns in stock prices.

B. Facebook Prophet

Another essential part of the web application for stock market prediction is Facebook Prophet, which provides an alternative method of forecasting to the LSTM model. Facebook's Core Data Science team developed Prophet, which is renowned for its versatility, ease of use, and ability to recognize a broad variety of time-series patterns, such as trends, seasonality, and holidays. Prophet is especially made for time-series forecasting activities. Prophet works by breaking down time-series data into components related to trend, seasonality, and holidays. These components are then used to model the data using additive regression models.This approach enables Prophet to generate reliable forecasts even with noisy and irregularly spaced data. In the context of the web application, Prophet provides an alternative method for predicting stock prices, allowing users to compare and contrast forecasts generated by both the LSTM model and Prophet. The integration of Prophet enhances the robustness and diversity of forecasting capabilities available to users, thereby enriching their decision-making process in the stock market.

C. Data Processing

The stock market prediction web application's data processing includes a thorough pipeline designed to convert unstructured historical stock market data into a format that can be analysed and modelled. Initially, data retrieval from external sources like the Yahoo Finance API provides access to pertinent stock market information such as price metrics and trading volume across various timestamps. Subsequently, a thorough data cleaning process ensues to rectify any inconsistencies, outliers, or missing values present in the dataset, ensuring data integrity. Following this, data transformation techniques are applied to standardize timestamps, adjust time zones, and format numerical data appropriately for further analysis. Feature engineering is then undertaken to extract relevant features, including moving averages and technical indicators, which serve as inputs to predictive models. Moreover, numerical features are normalized or scaled to a uniform range to improve machine learning models' performance. To aid in the evaluation of the model, the dataset is then divided into training and testing sets with a significant portion allocated for training and the remainder reserved for testing. Throughout this process, data visualization techniques such as line plots and histograms are utilized to glean insights into the underlying data distribution and identify patterns or trends. By systematically executing these steps, the data processing pipeline primes the raw stock market data for subsequent analysis and modeling within the web application, ultimately enabling users to make informed decisions based on accurate predictions of future stock prices.

D. Evaluation

Evaluation serves as a pivotal component within the stock market prediction web application, ensuring the reliability and efficacy of the predictive models deployed. This process encompasses several key facets aimed at assessing the models' performance and predictive accuracy.The dataset is first divided into subsets for testing and training, the latter of which is used to assess predicted performance on unseen data and the former for model training. The discrepancy between the actual and anticipated stock prices is then measured using a variety of performance indicators, such as R-squared (R2) score, Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE). Furthermore, cross-validation methods like k-fold cross-validation are applied to evaluate the robustness of the model and address problems like overfitting. By mimicking actual trading situations, backtesting—which involves assessing model performance using previous data—further improves evaluation.Visualizations such as line plots and time-series plots are leveraged to intuitively compare predicted and actual prices, enabling users to gain insights into model effectiveness and identify trends or patterns. Through these evaluation processes, the stock market prediction web application ensures that users can make well-informed

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decisions based on accurate forecasts of future stock price movements.

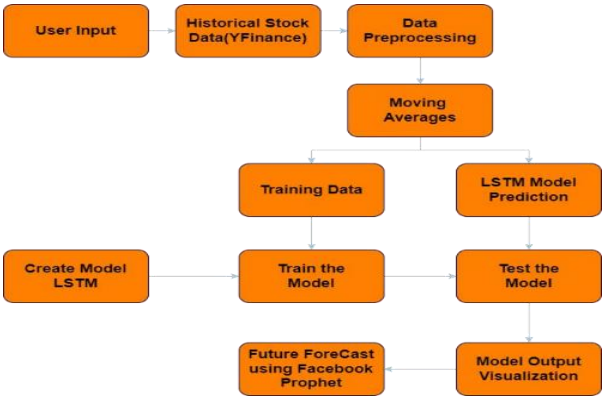

Fig. 1. Architecture Diagram

IV. Result & Discussion

The results and discussion section of the stock market prediction web application encompasses the analysis and interpretation of the model outputs, as well as the implications for investors and users. This section typically begins with a summary of the performance metrics obtained from evaluating the predictive models, including RMSE scores. . These metrics provide quantifiable measures of the models' accuracy and precision in fore casting future stock prices.

Following the presentation of performance metrics, the discussion delves into the strengths and limitations of the predictive models. This involves an examination of the factors contributing to model accuracy, such as the inclusion of relevant features, model complexity, and the quality of the training data. Additionally, any challenges or shortcomings encountered during model development and evaluation are addressed, along with potential strategies for improvement.

The discussion also explores the practical implications of the model outputs for investors and users of the web application. This includes insights into potential investment opportunities, risk management strategies, and the role of predictive models in supporting decision-making in the financial markets. Furthermore, the discussion may touch upon the broader implications of stock market prediction, including its impact on market efficiency, investor behavior, and regulatory considerations.Incorporate visual aids like time-series plots comparing actual vs. predicted stock prices, or heatmaps to illustrate feature importance. These visualizations can help users intuitively understand model performance and key drivers of predictions.Discuss how the application could provide actionable insights, such as personalized alerts for significant market trends or forecasts tailored to specific investment preferences, enhancing the application's usability and value for diverse user groups. Include a discussion of how the model performs under different market conditions, such as in various trends or volatile periods. This analysis helps users understand the robustness and reliability of predictions in varying economic climates.

Overall, the results and discussion section serves to provide a comprehensive analysis of the predictive models' performe, as well as their implications for stakeholders.

1. *Input the Stock Ticker*

Enter the Stock id

GOOGl

Fig. 2. Stock id entry

Table I. Google Stock Data(year:2023)

Date	Open	High	Low	Close
1/2/23	26.63	26.79	26.36	26.47
1/5/23	26.375	26 .389	25.887	25.93
1/6/23	26	26.077	25.5	25.732
1/7/23	25.547	25.747	25.277	25.42
1/8/23	25.72	25.96	25.53	25.9
1/9/23	25.93	26.34	25.505	25.76
1/12/23	25.77	25.845	25.36	25.41
1/13/23	25.58	25.64	25.41	25.45
1/14/23	25.25	25.477	25.2	25.36
1/15/23	25.61	25.93	25.375	25.46
1/16/23	25.55	25.92	25.48	25.62
1/20/23	25.44	25.487	25.102	25.205

Layer (type)	Output Shape	Param #
lstm_1 (LSTM)	(None, 100, 128)	66,560
lstm_2 (LSTM)	(None, 64)	49,408
dense (Dense)	(None, 25)	1,625
dense_1 (Dense)	(None, 1)	26

Total params: 352,859 (1.35 MB)

Trainable params: 117,619 (459.45 KB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 235,240 (918.91 KB)

```
predictions=model.predict(x_test)
```

46/46 ━━━━━━━━━━━━━━━━━━━━ 3s 57ms/step

Fig. 3. Train the LSTM model and predict the data

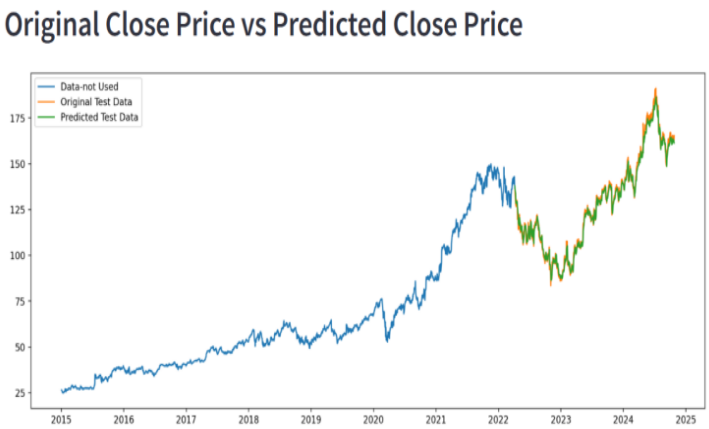

Fig. 4. Prediction using LSTM Model

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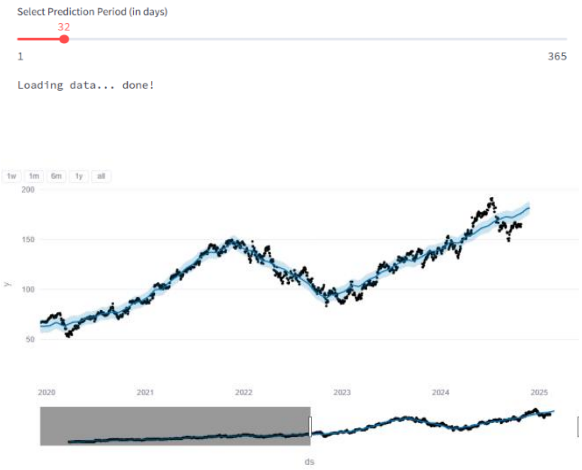

Fig. 5. Prediction using Facebook Prophet

V. Conclusion

The 'Stock Market Prediction Web App using LSTM Model and Facebook Prophet' project combines Facebook Prophet and LSTM models with the user- friendly interface of Streamlit to produce a reliable stock price prediction tool that benefits investors.

The project uses sophisticated predictive analytics to help users find intricate patterns and trends in historical stock market data, which helps with risk management and decision- making.

Users may easily access and interact with the programme from any internet-connected device thanks to the user-friendly online interface, which improves accessibility.

In addition to its usefulness, the project is an instructional tool that gives users the chance to learn more about the complexities of time series forecasting and machine learning. The project is positioned for ongoing adaptation and improvement, which keeps it relevant and successful in the always changing financial scene while providing analysts and investors with prospects for financial success and actionable information.

VI. FUTURE ENHANCEMENT

The "Stock Market Prediction Web App using LSTM Model and Facebook Prophet" project holds immense potential for future enhancements and expansion.

A. Integration of Other Data Sources:

Predictive models can be improved and more thorough insights into market behaviour can be obtained by incorporating data from a greater range of sources than only stock market data. Examples of these sources include social media trends, sentiment analysis in the news, and macroeconomic indicators.

B. Improved Visualisation Features:

Adding interactive and adaptable visualisation elements to the interface, including candlestick charts, heatmaps, and technical indicators, can improve user experience and make it easier to analyse stock market patterns in greater detail.

C. Advanced Analytics and Insights:

Offering advanced analytics and insights to help customers find special opportunities and patterns in the stock market that might not be visible by conventional analysis techniques, such as anomaly identification, trend analysis, and correlation analysis.

E. User Customisation and Personalisation:

Giving users the option to personalise the application to their unique investing plans and preferences, as well as to customise their dashboard and establish personalised alerts and notifications.

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